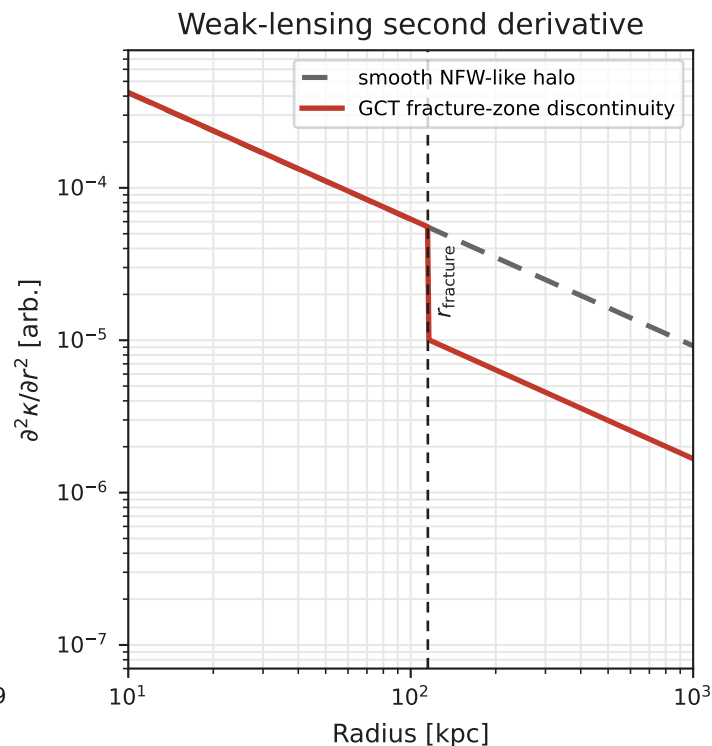
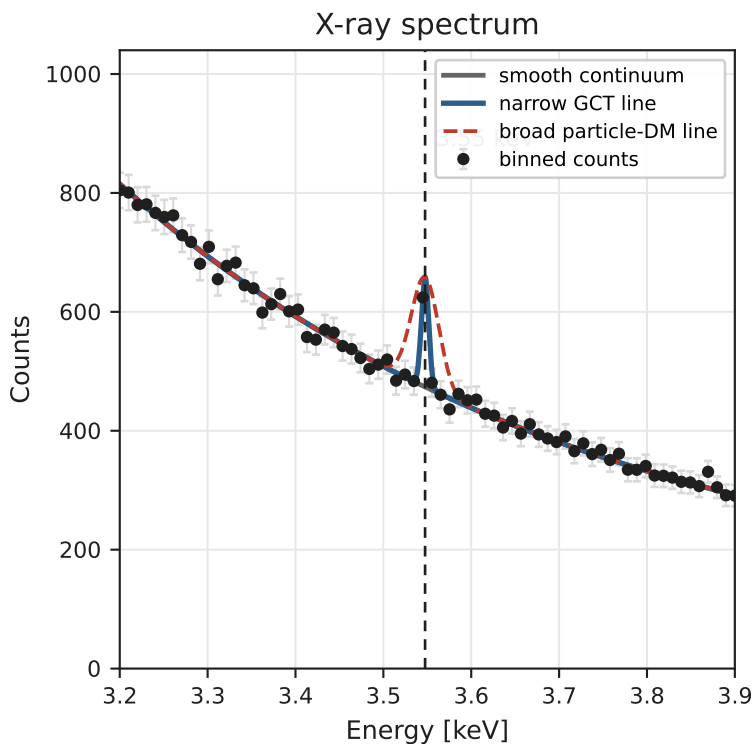


The 3.55 keV X-ray Anomaly: Detection and Morphology Tests

Perseus cluster context: line shape plus second-derivative lensing morphology.



XRISM line gate

GCT FWHM ~ 9.7 eV
CDM FWHM ~ 36.6 eV

Morphology gate

sharp lensing
second-derivative
feature

Null outcome

smooth halo +
no narrow residual

Observation:
line residual + lensing map

narrow line + discontinuity
supports fracture morphology

broad line or smooth halo
routes to particle-DM/null